

He-3 Neutron Detector

Type 70 063

REF 063 00 93

Application

The neutron detector 70 063 is a proportional counter tube filled with Helium-3 to provide maximum sensitivity to thermal or thermalized neutrons. It is nearly insensitive to gamma-radiation.

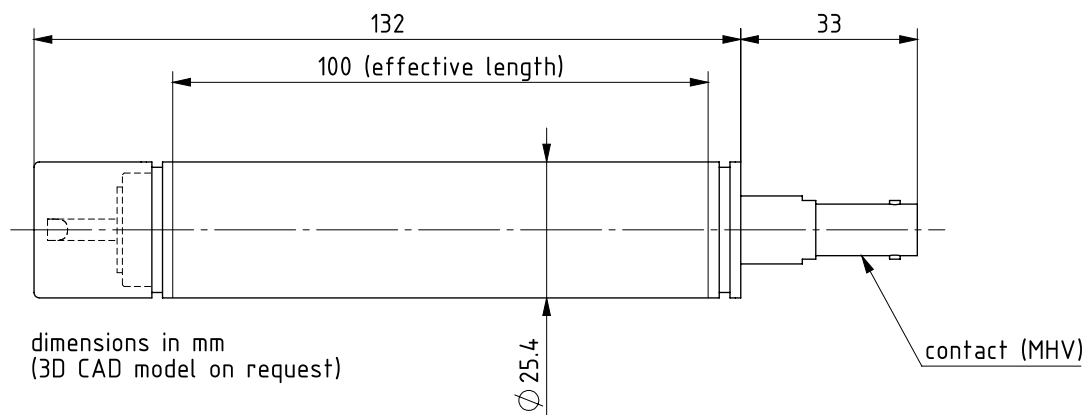
This neutron detector is suitable for portable and handheld instruments detecting neutrons, for the measurement of neutron flux densities, even in mixed neutron-gamma radiation fields, for neutron measurements in cavities or pipes, for safeguards, and for various other industrial and scientific applications.

The detector type 70 063 / order no. 063 00 93 with 1/1 inch external diameter and 100 mm effective length has a maximum total length of 165 mm. The gas pressure of 4 bar results in a sensitivity of 8.60 counts/s per neutron/cm²/s for thermal or thermalized neutrons.

Construction Type

The counter tube is made of stainless steel of the type ANSI 316 (EN 1.4435) with an alumina ceramic insulator as feed through. It possesses a MHV-connector with Teflon insulation. The thickness of the wall is 2.11 mm.

Mechanical Data



Measurement Circuit

The given data was measured by counting the pulses derived from the anode signal by a charge sensitive preamplifier, pulse processing, and multichannel pulse height analysis.

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Technical Data

All data refer to the recommended operation conditions.

General Data

Thermal neutron sensitivity	8.60 cps/nv $\pm 10\%$ *1,2,3)
Thermal neutron flux density range	$6 \cdot 10^{-3}$ to $1 \cdot 10^{+4}$ nv *4)
Background due to photon radiation	$< 10^{-5}$ counts/photon/cm ²
Detector count rate by terrestrial neutrons at earth surface	$\approx 5 \cdot 10^{+1}$ cph *5)
Resolution (broadening at 764 keV, FWHM)	$< 10\%$
Maximum diameter	1/1 inch (25.4 mm)
Maximum overall length	165 mm
Effective gas volume	35.2 cm ³
Length of effective gas volume	100 mm
Neutron sensitive material	³ He
Additive	CO ₂
Gas pressure	4 bar
Mass	200 g

Electrical data

Recommended operating voltage	760 V
Operating voltage for a gas multiplication factor of 10 to 100	630 to 890 V
Anode to cathode capacitance	≈ 4.6 pF
Insulation resistance	$> 10^{+8}$ M Ω

Limiting values

Maximum voltage	2000 V
Maximum temperature	100 °C

*1) The unit cps/nv is equivalent to counts/s per neutron/cm²/s.

*2) The thermal neutron sensitivity is based on measurements at the PTB Thermal Neutron Calibration Facility. The neutron reference field of this facility is built by the neutrons escaping from the surface of a graphite moderator at 30 cm distance.

*3) The uncertainty interval is for a confidence level of approximately 95 %.

*4) Minimum for 10 % statistical accuracy after 30 min, maximum for a pulse processing period < 10 μ s.

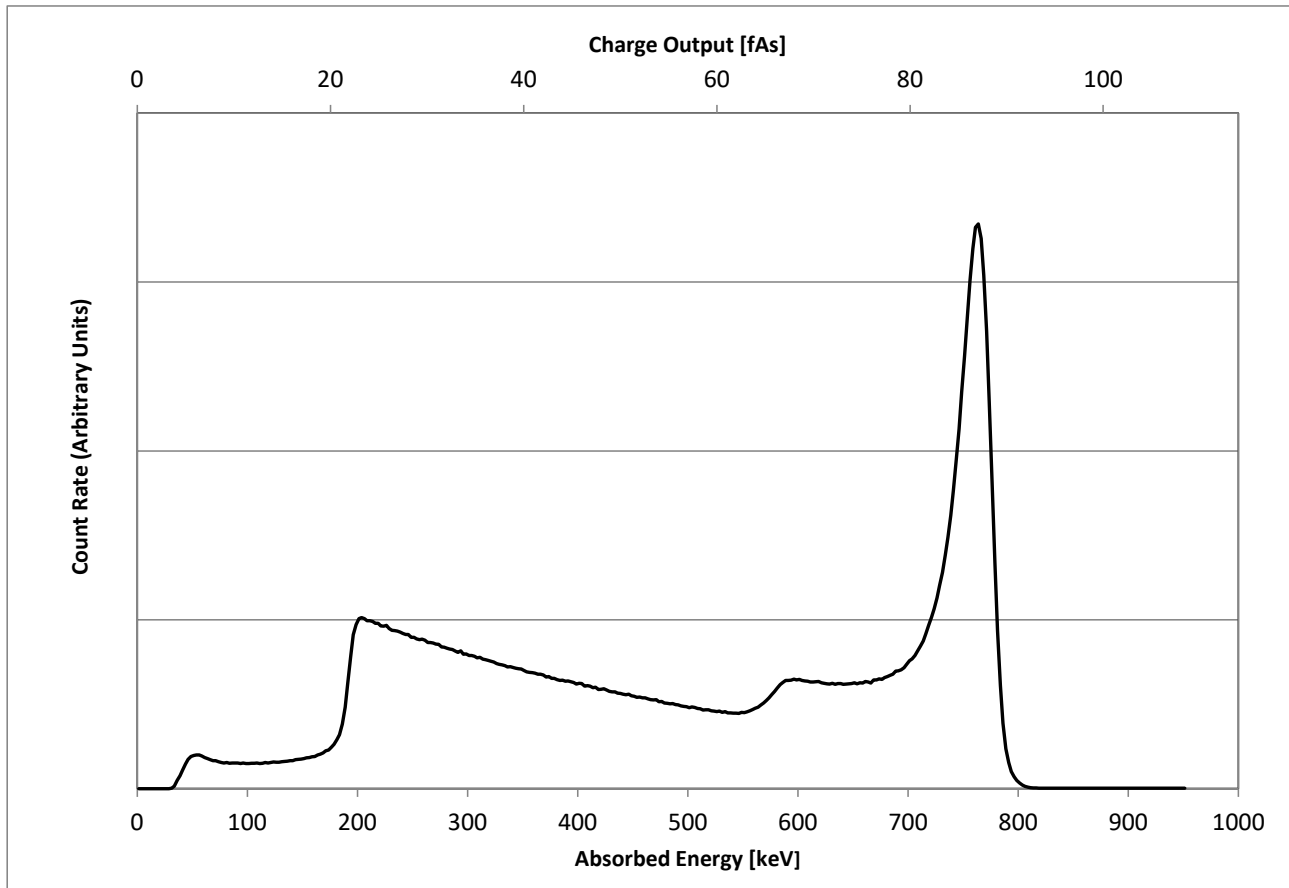
*5) This value is only a rough estimate for the expected count rate.



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Typical Pulse Height Spectrum



The pulse height spectrum has been measured with a moderated Cf-252 source, digital pulse shaping equivalent to 7 μ s Gaussian shaping, and the recommended operating voltage of 760 V. The charge output in the full absorption peak (764 keV) is $Q \approx (2.9 \times M)$ fAs with M being the actual multiplication factor of the proportional counter tube. The resolution in the pulse height spectrum depends on the multiplication factor M, the pulse shaping and the quality of the pulse processing electronics.